Cocu Matei-Iulian
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EDUCATION

2024 – 2027 *University of Bucharest*
Faculty of Mathematics and Computer Science
Specialization: Computer Science | Admission GPA: 9.93 | 1st year GPA: 9.28 | 2nd year GPA: 9.35
(top 10% of the class thoroughly)
Faculty of Psychology and Education Sciences – Pedagogical Training Program

2020 – 2024 *National College "Vasile Alecsandri", Galați*
Romanian Baccalaureate: 9.66

WORK EXPERIENCE

2026: **Backend SWE Apprenticeship**, *Code of Talent*. Completed a 2-month apprenticeship developing backend services using Symfony; implemented RESTful endpoints, business logic modules, and database query optimizations for production web applications.

2026: **Computer Science Tutor**, *Algorithm Tutoring Center*. Instructed high school seniors in advanced data structures, algorithmic problem-solving, and preparation for national examinations.

2025 – 2026: **Academic Peer Tutor**, University of Bucharest. Conducted weekly laboratory and seminar sessions in Introductory Computer Architecture and Relational Databases for cohorts of ∼15 first-year undergraduate students.

RESEARCH & PUBLICATIONS

Investigating Bias Patterns in LLMs Across Romanian Historical Contexts
Co-authored paper accepted at ConsiLR 2025: Analyzed socio-political and historical bias in Small/Large Language Models across sensitive Romanian historical topics using structured probing benchmarks and qualitative evaluation.

PROJECTS

CivicMind Co-developed an AI-driven platform for legislative transparency, aggregating public policy data to generate automated law summaries, Pro/Con analyses, and representative Impact Scores. Engineered end-to-end asynchronous data scrapers and REST APIs using FastAPI, Django, and PostgreSQL; integrated multi-agent LLM pipelines via LangGraph and Mistral APIs for structured text synthesis; integrated minimal frontend via native React.

Grapher – Task Management Web App Engineered a full-stack project tracking application using ASP.NET Core and C# on the backend with Mermaid.js for interactive DAG workflow visualization; implemented JWT authentication, task CRUD operations, and real-time state updates.

Signal Object Detection Trained and tuned CNN architectures in PyTorch for multi-class signal classification, handling dataset preprocessing, feature normalization, and metric-based hyperparameter optimization.

HLTV-DB Designed a normalized relational database schema modeling professional Counter-Strike esports analytics (inspired by HLTV.org); engineered optimized PL/SQL triggers, stored procedures, and complex analytical queries deployed on an Oracle Database instance.

QLTV – Esports Analytics Desktop Client Developed a desktop application client for the HLTV-DB system using modern Java 21 and JavaFX; implemented an MVC architecture with a native JDBC data access layer to execute real-time statistics queries, tournament bracket tracking, and player match analysis.

macoq – Concurrency Synchronization Library Built a lightweight, header-based C concurrency library utilizing C11 atomic operations (*<stdatomic.h>*) to implement thread-safe synchronization primitives, spinlocks, and lock-free queues without POSIX mutex dependencies.

QR Code Generator Implemented a standalone QR code generator from scratch in JavaScript; programmed data byte encoding, masking patterns, matrix placement rules, and Reed-Solomon error correction algorithms without third-party dependencies.

ToonLander – 2D Platformer Game Developed a modular 2D platformer engine in C++ and SFML applying OOP paradigms; built a decoupled game loop, frame-rate independent state manager and dynamic collision detection.

HONORS, AWARDS & ACTIVITIES

2024: Third Prize, MateInfoUB Competition – Mathematics & Informatics Sections.

2020 – 2023: Volunteer Member, Leo Alpha Danubius Galați. Managed outreach logistics and led community civic projects.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, Java, C#, PHP, JavaScript (TypeScript), SQL (PL/SQL), Haskell, x86_asm
- **Backend & Web:** FastAPI, Django, Symfony, ASP.NET Core, Spring Boot, React, RESTful APIs
- **AI, ML & Core:** PyTorch, JAX, LangGraph, Multithreading/Concurrency
- **Databases, DevOps & Other Misc.:** PostgreSQL, Oracle DB, Docker, Git, Linux/Bash, LaTeX

LANGUAGE SKILLS

Romanian: Native | **English:** Fluent / Bilingual (C2 Proficient, Cambridge CAE(CPE) Score: 202) | **German:** Elementary (B1)